



AI Powered Stock Recommender System

Project Proposal

Project Advisor:

Mr. Danyal Farhat

Group Members:

Bassam Khalid <22L-7456>

Rameez Rehan <22L-7502>

Abdur Rub Mustafa <22L-7538>

National University of Computer and Emerging Sciences
Department of Computer Science
Lahore, Pakistan

Abstract

This project proposes the development of an AI-based stock recommendation system that integrates sentiment analysis, technical indicators, and potentially fundamental analysis to provide actionable Buy, Sell, or Hold recommendations. A unique aspect of this project is its aim to test Eugene Fama's Efficient Market Hypothesis (1970), which argues that historical price data alone is sufficient to reflect all available information and maximize profit opportunities. By comparing models based solely on historical price data against those that also incorporate financial news sentiment, market bias, and fundamentals, the system will attempt to prove or disprove this long-standing theory.

Using advanced natural language processing (NLP) models such as FinBERT, transformer-based architectures (e.g., BERT, RoBERTa, GPT-based embeddings), and modern time-series deep learning models (LSTMs, GRUs, Temporal Convolutional Networks, and Transformers for sequential data), the project will evaluate which combination yields the most robust recommendations. Results will be delivered through a user-friendly web-based interface with an integrated AI chatbot, enabling retail investors to interactively query and receive real-time stock advice enriched with confidence scores and rationales.

1. Introduction

This proposal outlines the development of an AI-powered stock recommendation system that not only helps retail investors but also contributes to an academic test of the Efficient Market Hypothesis (EMH). Eugene Fama (1970) argued that markets are efficient and that all available information is already reflected in stock prices, implying that technical analysis of historical price data should be sufficient for investment decisions.

However, recent advances in artificial intelligence suggest that integrating sentiment analysis of financial news, social media, and fundamental indicators may provide an edge over purely historical data-driven approaches. This project therefore investigates whether AI models incorporating external signals can outperform models limited to historical stock data, thereby evaluating the validity of Fama's claim in the context of modern markets.

The system will not only generate Buy/Sell/Hold recommendations but will also provide explainability (through confidence scores, SHAP/LIME explanations) and an AI-powered chatbot interface to make results more accessible for non-expert retail investors.

2. Goals and Objectives

The objectives of this project are as follows:

1. To test Eugene Fama's Efficient Market Hypothesis (1970) by comparing models using only historical stock data with those integrating sentiment and/or fundamental data.
2. To automatically analyse market sentiment from financial news and social media using NLP models.

3. To integrate historical price movements with advanced technical indicators (Moving Averages, RSI, MACD, Bollinger Bands, etc.).
4. To explore the impact of fundamental financial metrics (e.g., P/E ratio, EPS, revenue growth) on prediction quality.
5. To implement modern AI/ML techniques for time-series and text analysis, including:
 - Transformer-based models
 - LSTMs and its fine-tuned models
 - Graph Neural Networks (GNNs) for relational stock dependencies
 - Reinforcement Learning for trading strategy optimization
 - Ensemble learning methods
6. To present actionable recommendations (Buy, Sell, Hold) with confidence scores and explanatory rationales.
7. To design and deploy a web-based UI with an integrated AI chatbot that can provide interactive recommendations

3. Scope of the Project

The scope of this project includes the following:

- Accepting user input for a stock (e.g., ticker symbol or company name).
- Retrieving historical price data and applying time-series models for prediction.
- Collecting financial news, social media posts, and fundamental company data.
- Performing **sentiment analysis** using state-of-the-art NLP models such as FinBERT, LLaMA 3 fine-tuned for finance, or HuggingFace transformer pipelines.
- Computing technical indicators and using them in combination with ML models.
- Building comparative pipelines:
 - Historical Data Only (Fama's hypothesis baseline)
 - Historical + Sentiment + Fundamentals + Technical Analysis
- Generating Buy, Sell, or Hold recommendations supported with rationale and evidence.
- Delivering outputs via a web interface (Streamlit/Flask) with an AI chatbot for conversational querying.

4. Initial Study and Work Done so Far

The initial study carried out includes:

- Reviewing Eugene Fama's Efficient Market Hypothesis (1970) and its implications for price prediction.
- Reading research articles related to stock market prediction, sentiment analysis, and hybrid AI models in finance.
- Exploring the S&P 500 index as a testbed for evaluation, due to its importance as a global benchmark.
- Studying existing work on combining technical indicators with sentiment and fundamental analysis.
- Identifying APIs and datasets such as Yahoo Finance, Alpha Vantage, Polygon.io, and financial news APIs.

References

- [1] H. Kaeley, Y. Qiao, and N. Bagherzadeh, “Support for Stock Trend Prediction Using Transformers and Sentiment Analysis,” *arXiv preprint*, arXiv:2305.14368, 2023.
- [2] S. Halder, “FinBERT-LSTM: Deep Learning Based Stock Price Prediction Using News Sentiment Analysis,” *arXiv preprint*, arXiv:2211.07392, 2022.
- [3] J. B. Smith and A. Gupta, “Drivers of S&P 500’s Profitability: Implications for Investment Strategy and Risk Management,” *Economies*, vol. 12, no. 4, p. 77, 2024.
- [4] E. F. Fama, “Efficient Capital Markets: A Review of Theory and Empirical Work,” *Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.
- [5] S. Omranpour, G. Rabusseau, and R. Rabbany, “Higher Order Transformers: Enhancing Stock Movement Prediction on Multimodal Time-Series Data,” *arXiv preprint*, arXiv:2412.10540, 2024.
- [6] *Deep reinforcement learning for portfolio selection*, *Global Finance Journal*, vol. 62, Article 101016, Sept. 2024.